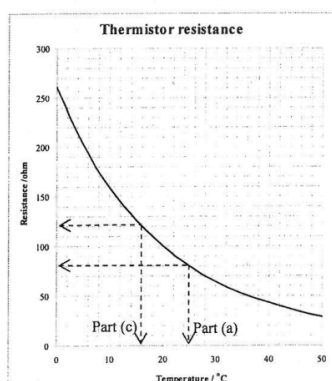


- |    |     |       |                                                                                                                                                                             |          |          |
|----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
| 3. | (a) | (i)   | $P = F v$<br>$= 8000 \text{ N} \times 2 \text{ m s}^{-1}$<br>$= 16 \text{ kW}$                                                                                              | 1M       |          |
|    |     |       |                                                                                                                                                                             | 1A       | <u>2</u> |
|    |     | (ii)  | $P_{\text{loss}} = 20 \text{ kW} - 16 \text{ kW} = 4 \text{ kW}$                                                                                                            | 1A       | <u>1</u> |
|    | (b) | (i)   | $P = 4 \text{ kW} + (8000 - 7000 \text{ N}) \times 2 \text{ m s}^{-1}$<br>$= 6000 \text{ W} = 6 \text{ kW}$                                                                 | 1M<br>1A | <u>2</u> |
|    |     | (ii)  | Output power required from motor is smaller.<br><u>Or</u> Force exerted by motor is smaller.                                                                                | 1A       | <u>1</u> |
|    |     | (iii) | No, the lift system could not work / the lift will fall<br>as slipping would occur / the cable cannot be fixed on the drum / the drum cannot exert a<br>force on the cable. | 1A<br>1A | <u>2</u> |

10. (a) (i)  $80\ \Omega$ 

$$(ii) \quad V_{AB} = \frac{120}{(80 + 120)} \times 12 = 7.2\text{ V}$$

- (b) As  $R_v$  and  $120\ \Omega$  resistor are in parallel,  $R_{eq}$  across  $AB$  is smaller than  $120\ \Omega$ , therefore voltage shared across  $AB$  is reduced / smaller than expected.

Use a voltmeter with resistance much larger than the resistance in that part of the circuit. (e.g.  $10\text{ M}\Omega$  in some digital voltmeter)

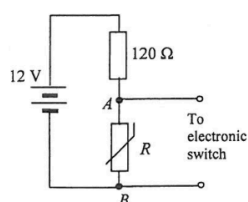
$$(c) (i) \quad V_{AB} = \frac{120}{(R + 120)} \times 12 = 6.0\text{ V}$$

$$R = 120\ \Omega$$

corresponds to temperature at  $16^\circ\text{C}$ .

58

10. (c) (ii) Correct circuit (i.e. interchange thermistor  $R$  and  $120\ \Omega$  resistor). As the temperature drops, the thermistor resistance increases. When the resistance increases to a value such that  $V_{AB} = 6.0\text{ V}$  or above, the electronic switch is on and it turns on the heating device.



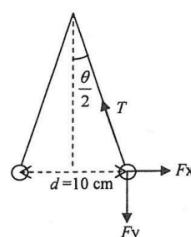
$$11. (a) \quad T \cos \frac{\theta}{2} = F_y = mg$$

$$T \sin \frac{\theta}{2} = F_x = \frac{Q^2}{4\pi \epsilon_0 d^2}$$

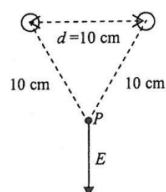
$$\tan \frac{\theta}{2} = \frac{Q^2}{4\pi \epsilon_0 d^2} \left( \frac{1}{mg} \right)$$

$$= 9 \times 10^9 \times \frac{(3.1 \times 10^{-9})^2}{0.1^2} \times \frac{1}{(1.0 \times 10^{-5})(9.81)}$$

$$\frac{\theta}{2} = 5.0^\circ \text{ i.e. } \theta = 10.1^\circ$$



- (b) (i)



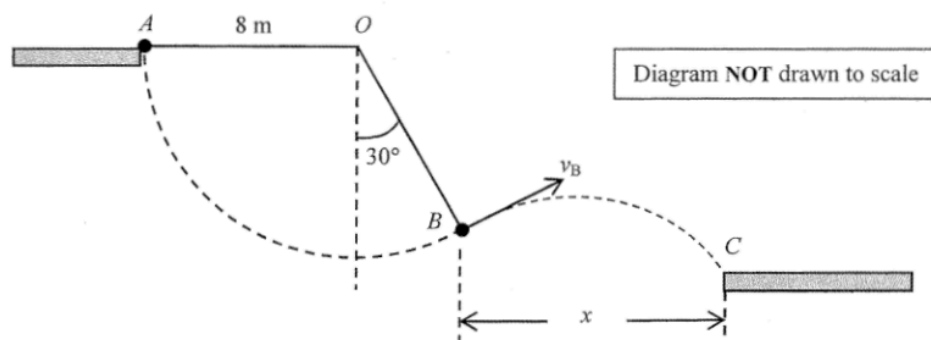
$$(ii) \quad \text{Potential at } P = \frac{Q}{4\pi \epsilon_0 d} + \frac{Q}{4\pi \epsilon_0 d} = \frac{2Q}{4\pi \epsilon_0 d}$$

$$= (9 \times 10^9) \frac{2 \times 3.1 \times 10^{-9}}{0.1}$$

$$= 558\text{ V}$$

- (iii) Separation  $d$  decreases.

3. (a)



$v_B$  correctly drawn with label (roughly perpendicular to  $OB$ ).

$$\frac{1}{2}mv^2 = mgh$$

$$v_B^2 = 2gh = 2 \times 9.81 \times (8 \cos 30^\circ)$$

$$v_B = 11.7 \text{ m s}^{-1} \text{ (11.65896) (or } 11.77 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})$$

(b) (i)  $x = v_x t = 11.7 \cos 30^\circ \times 1.25$  [ $v_x = v_B \cos 30^\circ$ ]  
 $= 12.6 \text{ m (12.62119)}$   
 (or 12.7 to 12.8 m for  $g = 10 \text{ m s}^{-2}$ )

(ii)

$$y = ut - \frac{1}{2}gt^2$$

$$u = v_y = v_B \sin 30^\circ = 5.83 \text{ m s}^{-1}$$

$$y = v_y(1.25) - \frac{1}{2}(9.81)(1.25)^2$$

$$y = -0.38 \text{ m (-0.414 to -0.352 m)}$$

$$\text{(or } -0.455 \text{ to } -0.4375 \text{ m for } g = 10 \text{ m s}^{-2})$$

Platform C is 0.38 m below B.

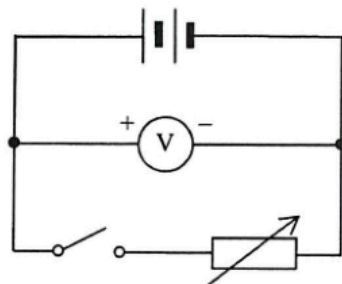
(c) Total mechanical energy is the same / unchanged.

|    |   |
|----|---|
| 1A |   |
| 1M |   |
| 1A |   |
|    | 3 |
| 1M |   |
| 1A |   |
|    | 2 |
| 1M |   |
| 1M |   |
| 1A |   |
|    | 3 |
| 1A |   |
|    | 1 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>7. (a) (i) <math>R = 10\text{ k}\Omega</math> (circuit I)</p> $V = \frac{\left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{10\text{ k}\Omega + \left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2\text{ V}$ <p><math>R = 100\text{ }\Omega</math> (circuit II)</p> $V = \frac{\left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{100\text{ }\Omega + \left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2.985\text{ V}$ | <p>1M</p> <p>1A</p> <p>1A</p> | <p>1M for appropriate method in calculating voltage</p> <p>Note: <math>100\text{ }\Omega</math> and <math>10\text{ k}\Omega</math> in parallel <math>\approx 99.0099\text{ }\Omega</math><br/>Accept stating that <math>V</math> slightly <math>&lt; 3\text{ V}</math></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                             |                                                                                                                                                                                                                                                                            |
| <p>(ii) Resistance of circuit / that part of circuit would be lowered / altered significantly when introducing the voltmeter (i.e. loading effect).</p> <p>Or The resistance of the voltmeter is comparable to the resistance of resistor <math>R</math>.</p>                                                                                                                                                                                                                                                                                                                     | <p>1A</p> <p>1A</p>           |                                                                                                                                                                                                                                                                            |
| <p>Resistance of voltmeter should be much higher than the resistance of the part of the circuit under study.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>1A</p>                     |                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                             |                                                                                                                                                                                                                                                                            |
| <p>(b) (i) <math>V_m</math> does NOT give the true voltage for the resistor.<br/><math>R_m = R_A + R</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>1A</p> <p>1A</p>           |                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                             |                                                                                                                                                                                                                                                                            |
| <p>(ii) For circuit III<br/><math>R_m = R + R_A = 10 + 1 = 11\text{ }\Omega</math><br/>percentage error <math>= \frac{1\text{ }\Omega}{10\text{ }\Omega} \times 100\%</math><br/><math>= 10\%</math></p>                                                                                                                                                                                                                                                                                                                                                                          | <p>1M</p> <p>1A</p>           |                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                             |                                                                                                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                                                                       |                                       |                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------|
| <p>3. (a) (i) (1) <math>\frac{1}{2}mv^2 = mgh</math><br/> <math>v^2 = 2(9.81)(12)</math><br/> <math>v = 15.344054 \text{ m s}^{-1} \approx 15.3 \text{ m s}^{-1}</math><br/> <math>(v = 15.491933 \text{ m s}^{-1} \approx 15.5 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})</math></p>     | <p>1M<br/><br/>1A<br/><br/>2</p>      | <p>Accept: (15.0 ~ 15.5) m s<sup>-1</sup></p> |
| <p>(2) <math>s = \frac{1}{2}gt^2</math><br/> <math>12 = \frac{1}{2}(9.81)t^2</math><br/> <math>t = 1.564124 \text{ s} \approx 1.56 \text{ s}</math><br/> <math>(t = 1.5491933 \text{ s} \approx 1.55 \text{ s} \text{ for } g = 10 \text{ m s}^{-2})</math></p>                                       | <p>1M<br/><br/>1A<br/><br/>2</p>      | <p>Accept: (1.50 ~ 1.60) s</p>                |
| <p>(ii) <math>F - mg = \frac{mv - mu}{t}</math><br/> <math>F = \frac{70 \times (0 - (-15.3))}{0.3} + 70 \times 9.81</math><br/> <math>= 4266.9793 \text{ N} \approx 4270 \text{ N}</math><br/> <math>(F = 4314.7845 \text{ N} \approx 4310 \text{ N} \text{ for } g = 10 \text{ m s}^{-2})</math></p> | <p>1M+1M<br/><br/>1A<br/><br/>3</p>   | <p>Accept: (4180 ~ 4320) N</p>                |
| <p>(iii) Elastic potential energy</p>                                                                                                                                                                                                                                                                 | <p>1A<br/><br/>1</p>                  |                                               |
| <p>(b) (i) (Velocity is too high, hence the force for deceleration is very large.)<br/>         - The life net may be torn.<br/>         - The falling person may be injured.<br/>         - The firemen may not be able to hold the life net tight.</p>                                              | <p>1A<br/><br/>Any ONE<br/><br/>1</p> |                                               |
| <p>(ii) There exists a horizontal velocity when a person jumps and the horizontal displacement is very difficult to estimate as it depends on the time of fall, which is relatively long.</p>                                                                                                         | <p>1A<br/>1A<br/><br/>2</p>           |                                               |

7. (a)



Close the switch and record corresponding  $V$  and  $R$  readings  
Adjust the resistance  $R$  to lower/other value(s) and repeat the experiment

Precaution:

- First set the variable resistor to its maximum / a large value
  - Open the switch after each measurement
  - Any reasonable answer
- } Any  
ONE

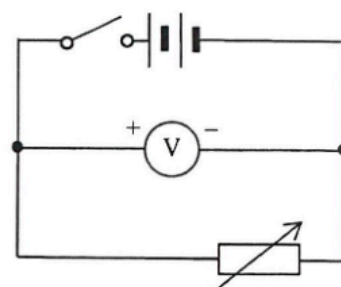
- (b) Terminal voltage  $V$  delivered increases with increasing (loading) resistance  $R$  (or graphical representation)

$$V = \xi \frac{R}{R+r} \quad \text{OR} \quad V = \xi - \frac{\xi}{R+r} r$$

1A  
1A

Correct circuit with correct symbol  
Correct polarity

Alternative circuit



1A  
1A

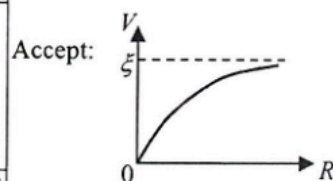
1A

5

1A

1A

2

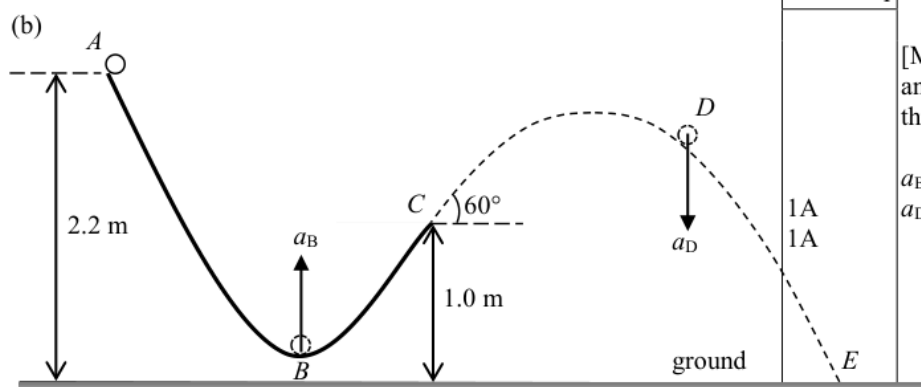


4. (a)  $v_B > v_C > v_D > v_A$

1A

[MR: Accept:  $v_A < v_D < v_C < v_B$   
Accept: B, C, D, A  
NOT accept: A, D, C, B]

1

1A  
1A

[MR: Accept the arrows drawn at anywhere on the vertical lines passing through the object at positions B & D]

$a_B$  correctly marked  
 $a_D$  correctly marked

- (c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and some kinetic energy is changed back to gravitational potential energy from B to C.

1A

2 [MR: 2 marks : All correct and complete: PE  $\rightarrow$  KE & some KE  $\rightarrow$  PE  
1 mark :

- (i) PE  $\rightarrow$  KE  $\rightarrow$  PE or  
(ii) Correctly describe only between 2 points (A  $\rightarrow$  B / B  $\rightarrow$  C / A  $\rightarrow$  C)]

2

- (ii)  $\frac{1}{2}mv^2 = mgh$   
 $v^2 = 2(9.81)(2.2 - 1.0)$   
 $v = 4.852216 \text{ m s}^{-1} \approx 4.85 \text{ m s}^{-1}$

1M  
1A

[MR:  $h$  must be correct for 1M  
Method using  $v^2 - u^2 = 2as$  etc.  
is NOT accepted]

For  $g = 10 \text{ m s}^{-2}$ ,  
 $v = 4.89898 \text{ m s}^{-1} \approx 4.90 \text{ m s}^{-1}$   
Accept  $4.85 \text{ m s}^{-1}$  to  $4.90 \text{ m s}^{-1}$

2

- (iii) Horizontal speed (at C)  $= 4.85 \times \cos 60^\circ$   
 $= 2.426108 \text{ m s}^{-1} \approx 2.43 \text{ m s}^{-1}$

1M

e.c.f. from (c)(ii)

$$\text{Distance} = 2.43 \times t = 2.55 \text{ m}$$

$$t = 1.051066 \text{ s} \approx 1.05 \text{ s}$$

1M  
1A

For  $g = 10 \text{ m s}^{-2}$ ,  
Horizontal speed  
 $= 2.44949 \text{ m s}^{-1} \approx 2.45 \text{ m s}^{-1}$   
**Vertical speed  $\approx 4.24 \text{ m s}^{-1}$**   
and  $t = 1.041033 \text{ s} \approx 1.04 \text{ s}$   
Accept  $1.04 \text{ s}$  to  $1.10 \text{ s}$

Or

$$\text{Vertical speed (at C)} = 4.85 \times \sin 60^\circ$$

$$= 4.2002 \text{ m s}^{-1} \approx 4.20 \text{ m s}^{-1}$$

$$-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$$

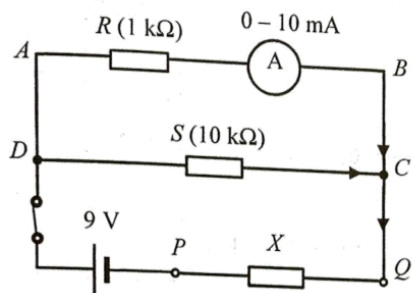
$$t = 1.051066 \text{ s} \approx 1.05 \text{ s}$$

[MR:  
1<sup>st</sup> M for correct expression for initial horizontal /vertical speed

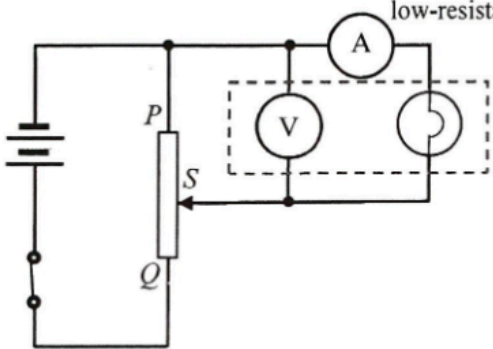
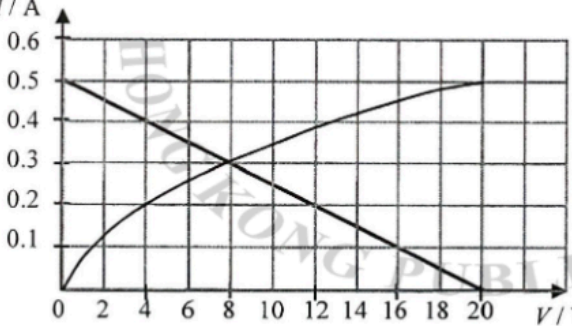
2<sup>nd</sup> M for correct method (equation) to find  $t$  by considering horizontal motion/ vertical motion

3

63

|                                                                                                                                                                                                                                                                                           | Marks              | Remarks                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. (a) (i) $V_{AB} = (8.5 \times 10^{-3} \text{ A}) \times (1 \text{ k}\Omega)$<br>$= 8.5 \text{ V}$                                                                                                                                                                                      | 1M<br>1A           |                                                                                                                                                           |
| (ii) Current through $S = \frac{8.5 \text{ V}}{10 \text{ k}\Omega}$<br>$= 8.50 \times 10^{-4} \text{ A} = 0.85 \text{ mA}$                                                                                                                                                                | 2<br><br>1M<br>1A  | As $AB$ and $CD$ are connected in parallel, current through $S$<br>$= (8.5 \times 10^{-3} \text{ A}) \times \frac{1 \text{ k}\Omega}{10 \text{ k}\Omega}$ |
| (iii)                                                                                                                                                                                                   | 2A<br><br>2        |                                                                                                                                                           |
| (iv) p.d. across $X = 9 - V_{AB} = 9 - 8.5 = 0.5 \text{ V}$<br>Current through $X = (8.5 \times 10^{-3}) + (8.5 \times 10^{-4})$<br>$= 9.35 \times 10^{-3} \text{ A}$<br>Resistance of $X = \frac{0.5 \text{ V}}{9.35 \times 10^{-3} \text{ A}}$<br>$= 53.476 \Omega \approx 53.5 \Omega$ | 1A<br>1M<br><br>1A |                                                                                                                                                           |
| (b) To prevent the ammeter from overloading (when $PQ$ is shorted or the resistance to be measured is too small)                                                                                                                                                                          | 3<br><br>1A<br>1   |                                                                                                                                                           |

|                                                                                                                                                                                                  |                      |                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------|
| 4. (a) $\frac{1}{2}mv^2 = mgh$<br>$= (50)(9.81)(1.5)$<br>$= 735.75 \text{ J} \approx 736 \text{ J}$                                                                                              | 1M<br>1A<br>2        | For $g = 10 \text{ m s}^{-2}$ , K.E. = 750 J |
| (b) (i) Kinetic energy and potential energy (of the athlete) change to elastic potential energy (of the trampoline).                                                                             | 2A<br>2              |                                              |
| (ii) $\bar{F}d = mgh + mgd$ [Or $\bar{F}d = \frac{1}{2}mv^2 + mgd$ ]<br><br>$\bar{F} = \frac{50(9.81)(1.5+0.40)}{0.40}$<br>$= 1839.375 + 490.5$<br>$= 2329.875 \text{ N} \approx 2330 \text{ N}$ | 1M<br><br><br><br>1A |                                              |
| Or<br>$\bar{F} = \frac{1}{0.40} [735.75 + (50)(9.81)(0.40)]$<br>$\approx 2330 \text{ N}$                                                                                                         | 1M<br>1A<br>2        |                                              |

| Solution                                                                                                                                                                                                                                                                                                                                            | Marks                            | Remarks                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------|
| 8. (a) (i)                                                                                                                                                                                                                                                         | low-resistance ammeter<br><br>1A |                                                        |
| (ii) Brightness increases.                                                                                                                                                                                                                                                                                                                          | 1A                               |                                                        |
| (b) (i) Resistance = $\frac{V}{I} = \frac{20}{0.5} = 40 \Omega$                                                                                                                                                                                                                                                                                     | 1M<br>1A                         |                                                        |
| (ii) As the applied voltage $V$ increases, current $I$ / electrical power increases and this raises the filament's temperature, thus the resistance $R$ of the filament / bulb increases.<br><u>Or</u><br>the electrons collide with the increasingly vibrating atoms / lattice ions (of the filament) which impede their flow, i.e. $R$ increases. | 1A<br>1A<br>1A                   |                                                        |
| (c) (i) <br>Current = 0.3 A (at $V = 8$ V)                                                                                                                                                                                                                       | 1M<br>1A                         | Correct straight line intersects the curve at (8, 0.3) |
| (ii) Power consumed = $IV$<br>$= (0.3)(8)$<br>$= 2.4 \text{ W}$                                                                                                                                                                                                                                                                                     | 1M<br>1A                         | e.c.f. from (c)(i)                                     |

|    |     |                                                                                                                               |    |   |
|----|-----|-------------------------------------------------------------------------------------------------------------------------------|----|---|
| 7. | (a) | $R = \rho \frac{l}{A} \Rightarrow 0.50 = \rho \frac{0.20}{8.0 \times 10^{-7}}$ $\rho = 2.0 \times 10^{-6} (\Omega \text{ m})$ | 1M |   |
|    |     |                                                                                                                               | 1A |   |
|    |     |                                                                                                                               |    | 2 |
|    | (b) | (i) Terminal 3                                                                                                                | 1A |   |
|    |     |                                                                                                                               |    | 1 |
|    |     |                                                                                                                               |    |   |
|    |     | (ii) When switch $S$ is connected to terminal 2, the equivalent resistance across $MN$                                        | 1M |   |
|    |     |                                                                                                                               | 1A |   |
|    |     |                                                                                                                               |    | 2 |
|    |     | (iii) (p.d. across $R_2$ is) greater than (p.d. across $R_1$ )                                                                | 1A |   |
|    |     |                                                                                                                               |    | 1 |
|    |     |                                                                                                                               |    |   |
|    |     | (iv) zero<br>as no p.d. across $CF$ / no current in the coil.                                                                 | 1A |   |
|    |     |                                                                                                                               | 1A |   |
|    |     |                                                                                                                               |    | 2 |

|         |                                                                                                                                                                                                                                                                                                                                                                         |          |                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------|
| 11. (a) | As the foil becomes longer / $l$ increases while its cross-sectional area becomes smaller / $A$ decreases due to decrease in width $w$ , its resistance would increase according to $R = \frac{\rho l}{A}$ .                                                                                                                                                            | 1A       |                                             |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 1A       |                                             |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 2        |                                             |
| (b) (i) | $V_{in} = I_1(R_1 + R_2)$                                                                                                                                                                                                                                                                                                                                               | 1A       |                                             |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 1        |                                             |
| (ii)    | Before:<br>$\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{500}{470 + 500} - \frac{470}{470 + 470} = 0.0154639$ After:<br>$\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{501}{470 + 501} - \frac{470}{470 + 470} = 0.0159629$ Thus $\frac{V_{out}}{V_{in}}$ increases $3.22687\% \approx 3.23\%$ . | 1M       |                                             |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 1A       | Accept $\pm(3.2 \sim 3.23)\%$ as the change |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 2        |                                             |
| (iii)   | (I) crack ①<br>(II) along $AB$                                                                                                                                                                                                                                                                                                                                          | 1A<br>1A |                                             |
|         |                                                                                                                                                                                                                                                                                                                                                                         | 2        |                                             |

|     |                                                                                                                                               |                                                                                                                                      |    |                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.  | (a)                                                                                                                                           | $s = ut + \frac{1}{2}at^2$ $0 = 5t + \frac{1}{2}(-9.81)t^2$ $t = 1.0194 \text{ s} \approx 1.02 \text{ s or } 0 \text{ s (rejected)}$ | 1M | <div style="border: 1px solid black; padding: 5px; width: fit-content;">           Or<br/> <math>v = u + at,</math><br/> <math>0 = 5 + (-9.81)(t/2)</math><br/> <math>t = 1.02 \text{ s}</math> </div> |
|     |                                                                                                                                               |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 2  |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1M |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
| (b) | speed of the bead at its maximum height = $2 \text{ m s}^{-1}$<br>$\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$ $= 0.02 \text{ J}$       |                                                                                                                                      | 2  | Accept: $t = 1.00 \sim 1.02 \text{ s}$                                                                                                                                                                 |
|     |                                                                                                                                               |                                                                                                                                      | 1M |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 2  |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1M |                                                                                                                                                                                                        |
| (c) | The bead and the trolley have the same horizontal velocity and thus travel the same horizontal distance (within the same time interval $t$ ). |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 2  |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 1A |                                                                                                                                                                                                        |
|     |                                                                                                                                               |                                                                                                                                      | 2  |                                                                                                                                                                                                        |

|     |                                                                                                                                                                                                                      |                                                                                                         |    |                                                                                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------|
| 7.  | (a)                                                                                                                                                                                                                  | To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat. | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1  |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1M |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
| (b) | $R = \frac{V}{I} = \frac{3.5}{0.7}$ $= 5.0 \Omega (> 4.7 \Omega)$ The extra resistance ( $0.3 \Omega$ ) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply. |                                                                                                         | 3  |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1M |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 2  |                                                                                                                        |
| (c) | $I = 0.25 \text{ A at } l = 30 \text{ cm}$ $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$ $= 9.0 \Omega$                                                                                                               |                                                                                                         | 1M | Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$<br>e.c.f. from (b)<br>Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$ |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 2  |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 1A |                                                                                                                        |
|     |                                                                                                                                                                                                                      |                                                                                                         | 2  |                                                                                                                        |

$$= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$$

$$(ii) \quad E_{out} = (680 \times 10^6) \times (24 \times 3600) \\ = 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$$

$$(iii) \quad \text{Efficiency} = \frac{E_{out}}{E_{in}} \times 100\% \\ = \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\% \\ = 40.8\% \approx 41\%$$

$$(b) \quad (i) \quad P = VI \\ 680 \text{ MW} = 400 \text{ kV} \times I \\ I = 1700 \text{ A} = 1.7 \text{ kA}$$

$$(ii) \quad \text{Power loss} = I^2 R \\ = (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3) \\ = 13410 \text{ W} \approx 13.4 \text{ (kW)}$$

(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.

|    |   |
|----|---|
| 1M | 2 |
| 1A |   |
| 2  |   |
| 1M |   |
| 1A |   |
| 2  |   |
| 1M |   |
| 1A |   |
| 2  |   |
| 1A |   |
| 1A |   |
| 2  |   |

e.c.f. from (a)(i)(ii)

Accept: 41% ~ 42%

e.c.f. from (b)(i)